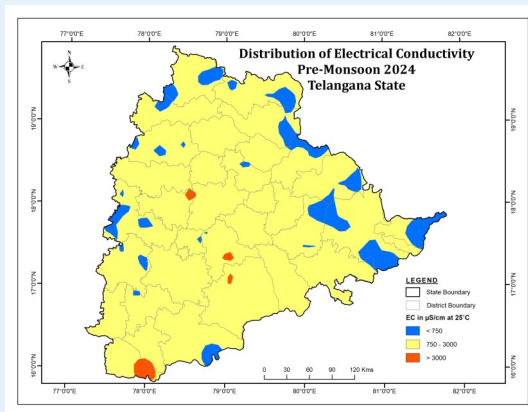


Groundwater Quality Scenario in Telangana

Parameters	No of samples	Permissible limit	No. of Samples above permissible limit	% Samples above permissible Limit
EC	1114	3000 $\mu\text{S}/\text{cm}$	30	2.69
Fluoride	1114	1.5 mg/L	167	14.99
Nitrate	1114	45 mg/L	290	26.03



Districts with anomalous values at sporadic locations

EC (3000 $\mu\text{S}/\text{cm}$)

Bhadradi Kothagudem, Jangaon, Jayashankar Bhupalapally, Jogulamba Gadwal, Kb Asifabad, Khammam, Mancheria, Medak, Nalgonda, Peddapalli, Ranga Reddy, Sangareddy, Siddipet, Warangal, Yadadri Bhuvanagiri

Fluoride (F > 1.5 mg/L)

Adilabad, Bhadradi Kothagudem, Hanamkonda, Hyderabad, Jagtial, Jangaon, Kamareddy, Karimnagar, Kb Asifabad, Khammam, Mahabubabad, Mahabubnagar, Medak, Medchal Malkajgiri, Nagarkurnool, Nalgonda, Nirmal, Nizamabad, Peddapalli, Rajanna Sircilla, Ranga Reddy, Sangareddy, Siddipet, Suryapet, Vikarabad, Wanaparthi, Warangal, Yadadri Bhuvanagiri

Nitrate (Nitrate > 45 mg/L)

Adilabad, Asifabad, Bhadradi Kothagudem, Hanamkonda, Jagtial, Jangaon, Jayashankar Bhupalapally, Jogulamba Gadwal, Kamareddy, Karimnagar, Kb Asifabad, Khammam, Mahabubabad, Mahabubnagar, Mancheria, Medak, Medchal Malkajgiri, Mulugu, Nagarkurnool, Nalgonda, Narayanpet, Nirmal, Nizamabad, Peddapalli, Rajanna Sircilla, Ranga Reddy, Sangareddy, Siddipet, Suryapet, Vikarabad, Wanaparthi, Warangal, Yadadri Bhuvanagiri

Arsenic (As > 10 ppb)

Not Any

Uranium (U > 30 ppb)

Not Any

For Further Information, Contact to :

Chairman, CGWB, Bhujal Bhawan,
NH IV Faridabad, Haryana - 121001



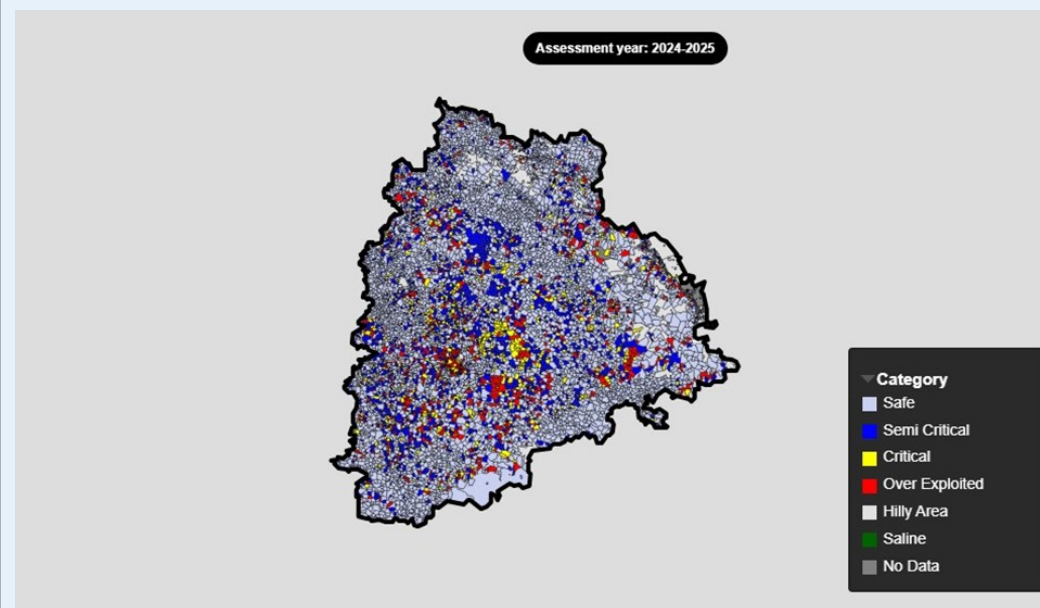
<https://ingres.iith.ac.in>

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Central Ground Water Board
Department of Water Resources, RD & GR
Ministry of Jal Shakti, Government of India



Dynamic Ground Water Resources, Ground Water Level &
Ground Water Quality of Telangana, 2025

November, 2025

Background

- Ground Water Resources Assessment (GWRA)- jointly carried out by Central Ground Water Board and State Nodal/Ground Water Department periodically as per the Ground Water Resource Estimation Committee (GEC) methodology.
- Carried out under the guidance of the respective State/UT Level Committees (SLCs) and overall supervision of Central Level Expert Group (CLEG).
- As part of the assessment, 'Annual Extractable Ground Water Resource' as well as 'Annual Ground Water Extraction' are assessed for each assessment unit (Mandal).
- The 'Stage of Ground Water Extraction' is computed as the ratio of 'Annual Ground Water Extraction' with respect to 'Annual Extractable Ground Water Resource' and is usually expressed in percentage. Based on the stage of extraction, the assessment units are categorized as Safe ($\leq 70\%$), Semi-Critical ($>70\%$ and $\leq 90\%$), Critical ($>90\%$ and $\leq 100\%$) and Over-Exploited ($>100\%$).
- GWRA- 2025, 2024, 2023, 2022 and 2020 has been carried out through a software/web-based application "INDIA-GROUNDWATER RESOURCE ESTIMATION SYSTEM (IN-GRES)" developed by CGWB through IIT-Hyderabad.

Salient Features

1	Rainfall	923.84 mm
2	Hydrogeology	Nearly 85 % of the State is underlain by hard rocks. Rest of the State is underlain by semi-consolidated formations and unconsolidated sediments.
3	Recharge Worthy Area of the State	1.06 Lakh Sq. Km
4	Assessment Unit (AU) Type / Number	Mandal / 620 Numbers
5	Average area of Assessment Unit	170.61 Sq. Km

Findings

	Attribute	GWRA-2017	GWRA-2020	GWRA-2022	GWRA-2023	GWRA-2024	GWRA-2025
1	Total Annual Ground Water Recharge (in bcm)	13.62	16.63	21.27	23.14	20.40	21.93
2	Annual Extractable Ground Water Resources (in bcm)	12.37	15.03	19.25	20.92	18.44	19.84
3	Annual Ground Water Extraction (in bcm)	8.09	8.01	8	8.09	8.47	9.26
4	Stage of Ground Water Extraction (in %)	65.45	53.32	41.6	38.65	45.91	46.69

bcm: Biliion Cubic Meters

Categorization of Assessment Units based on the 'Stage of Ground Water Extraction'

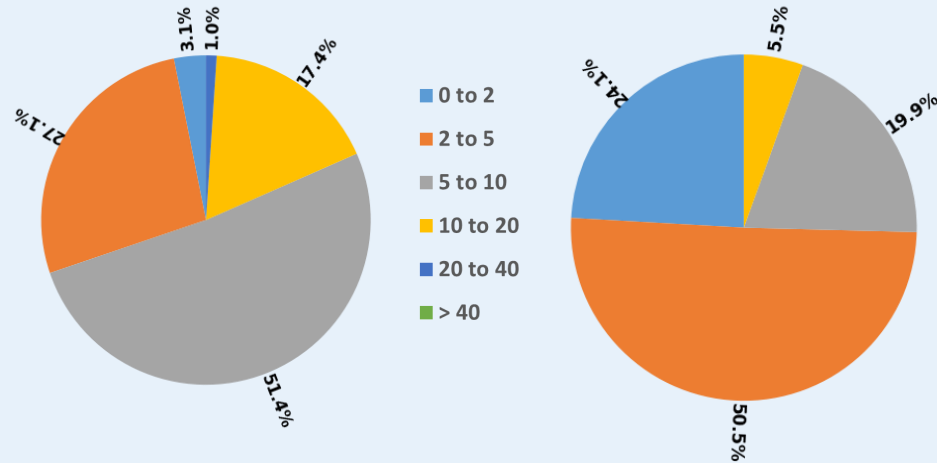
S. No	Category	GWRA-2017		GWRA-2020		GWRA-2022		GWRA-2023		GWRA-2024		GWRA-2025	
		Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs	Num-ber of AUs	% of AUs
1	Safe	278	48	321	55	494	83	530	86.60	490	79.03	473	76.29
2	Semi-critical	169	29	180	31	80	14	61	9.97	85	13.71	105	16.94
3	Critical	67	11	44	7	7	1	10	1.63	13	2.10	19	3.06
4	Over-exploited	70	12	44	7	13	2	11	1.80	32	5.16	23	3.71
5	Saline												
Total number of AUs		584		589		594		612		620		620	

Recommendations:-

- Telangana state is characterized by wide range of geological formations from Archaean to Recent age. Nearly 85% of the state is underlain by hard rocks (consolidated formations) belonging to the Peninsular Gneissic Complex, Dharwar and Eastern Ghats of Archaean to Middle Proterozoic age, Pakhal Group of rocks belonging to Middle to Upper Proterozoic age and Deccan Traps. Remaining of the state is underlain by semi consolidated sedimentary formations encompassing Gondwanas, Tertiary group of formations and Sub-Recent to Recent unconsolidated sediments.
- The Ground water resources for the state have been assessed watershed-wise and apportioned to mandal-wise. Total Annual Groundwater recharge of the State has been assessed as 21.93 bcm and Annual extractable Ground Water resource as 19.84 bcm. The Annual Ground Water Extraction is 9.26 bcm and Stage of Ground Water Extraction is 46.69 %. Out of 620 assessment units (Mandals), 23 units (4%) have been categorized as 'Over Exploited', 19 units (3 %) as 'Critical', 105 units (17 %) as 'Semi-Critical' and 473 units (76 %) as 'Safe'. There is no 'Saline' category of assessment unit in the state.
- More numbers of Water Harvesting and Conservation Structures may be constructed to catch the rain as the State is blessed with more than 900 mm annual rainfall particularly in the hard rock terrain. State may also effectively use "Master plan for Artificial Recharge" prepared by CGWB in consultation with State Government. (<https://cgwb.gov.in/cgwbprnm/publication-detail/324>).
- Restoration/rejuvenation of all the existing tanks should be taken up with the view of accommodating the available surface run off and thus augmentation of the ground water resources by artificial recharge. Periodical maintenance of these tanks is to be ensured. The "Manual on Artificial Recharge Techniques for augmentation of ground water" prepared by CGWB may be used for planning. (<https://cgwb.gov.in/sites/default/files/MainLinks/Manual-Artificial-Recharge.pdf>).
- Increase in irrigation efficiency through adopting of micro-irrigation techniques in more areas.
- Creating awareness (Mass Awareness Campaign for public and farmers, slideshows, display boards on water conservation, Water Management Training Programme for personnel related with water sector, painting/essay competition for school students etc.) regarding water conservation etc may be organized at appropriate level.
- State may review their free/subsidized electricity policy to farmers (if applicable), bring suitable water pricing policy and may work further towards crop rotation/diversification/other initiatives to reduce overdependence on groundwater.
- In Industrial areas, Disposal of industrial effluents, solid waste and urban sewerage should be disposed offsafely after treatment, so that the phreatic aquifer does not get adversely polluted.
- Regulation & control of Ground water Extraction: Ministry of Jal Shakti has issued the guidelines for control and regulations of ground water extraction vide notification dated 24.09.2020 which has further been amended in March 2023. Concerned departments may ensure implementations of the guidelines.

Groundwater Level Scenario in Telangana

1. Depth to Groundwater Level

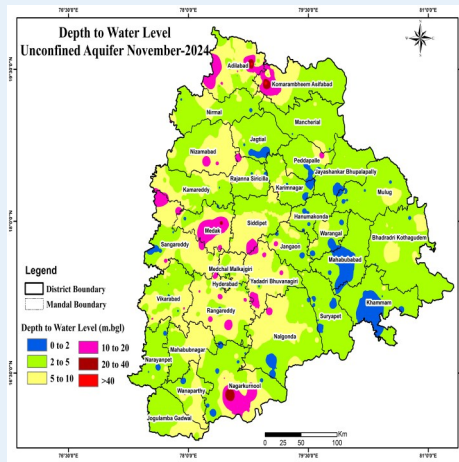
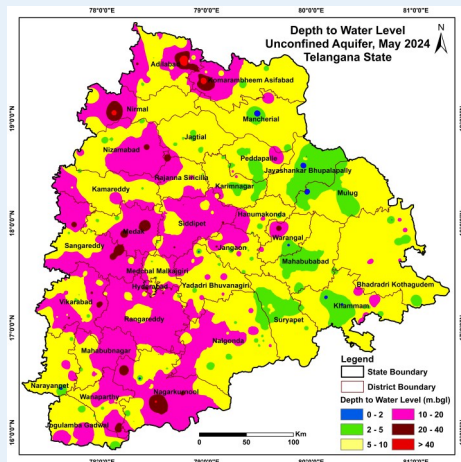


Pre-Monsoon (May 2024)

About 2% of wells recorded groundwater levels in 0–2 m bgl, 21% in 2–5 m bgl, 41% in 5–10 m bgl, 31% in 10–20 m bgl, 4% in 20–40 m bgl, and 1% in > 40 m bgl, with overall levels ranging from 0.48 m to 102 m bgl, indicating moderate to localized deep groundwater conditions.

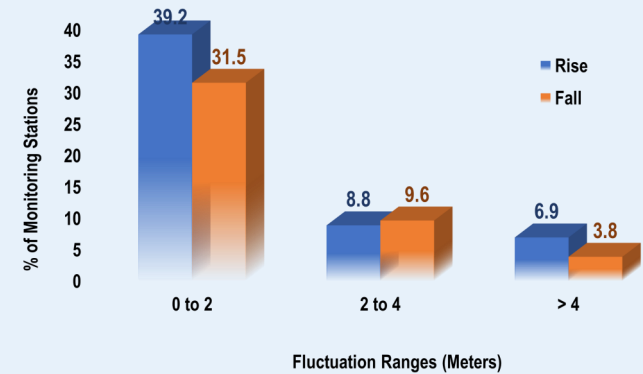
Post-Monsoon (November 2024)

About 19% of wells recorded groundwater levels in 0–2 m bgl, 29% in 2–5 m bgl, 30% in 5–10 m bgl, 16% in 10–20 m bgl, 4% in 20–40 m bgl, and 2% in >40 m bgl, with overall levels ranging from 0.1 m to 50.1 m bgl, indicating moderate to localized deep groundwater conditions.



2. Decadal Fluctuation of Groundwater Levels

Pre-Monsoon (May 2024 vs Decadal Mean: May 2014–2023)



Post-Monsoon (November 2024 vs Decadal Mean: November 2014–2023)

